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1. Executive Summary and Brand Direction

DEYOUNG COMMUNICATION is an AI communications platform that lets individuals, businesses, and agencies hire AI employees: voice and text agents that answer calls, hold natural conversations, execute approved tools, remember customers, and hand conversations to humans when the situation calls for it. This dossier is the Phase 1 planning package produced before implementation. It consolidates twenty-four required deliverables, from market research and competitor analysis through the product requirements document, technical specification, system and worker architecture, voice architecture, database schema, API design, permission and consent models, UX sitemap, screen inventory, design system, 3D and motion proposal, deployment and cost strategy, risk register, and implementation roadmap. Every decision in this document is written to survive contact with real engineering: no capability is claimed unless a production-grade path exists for it, and no number is presented unless it comes from a cited public source or an explicit engineering budget.

Brand Name Decision

The working name DEYOUNGCOMMUNICATION was evaluated against five public-facing variants: DEYOUNG COMMUNICATION, DEYOUNG COMM, DEYOUNG CONNECT, DEYOUNG VOICE, and DEYOUNG COMMUNICATIONS. The recommended public-facing brand is DEYOUNG COMMUNICATION, set as two words with a wordmark lockup. The reasoning is documented below rather than decided casually, as the master brief requires.

Candidate	Assessment	Verdict
DEYOUNG COMMUNICATION	Reads as a company about the act of communicating, which matches the product philosophy of selling communication rather than AI. Clean possessive-free construction, strong wordmark rhythm, and natural domain variants (deyoungcommunication.com, deyoungcomm.com).	Selected
DEYOUNG COMMUNICATIONS	The plural is the telecom-industry default, so it reads as infrastructure-first and blends in with thousands of legacy telcos and PBX vendors. It also invites the DEYOUNG COMM abbreviation, which weakens the brand.	Rejected
DEYOUNG CONNECT	Short and app-like, but generic. Dozens of connectivity products already use Connect and it discards the equity of the full family name.	Rejected
DEYOUNG VOICE	Accurately names the flagship capability, but it anchors the company to voice only and conflicts with the planned multi-channel roadmap (chat, SMS, email, WhatsApp).	Rejected

Candidate	Assessment	Verdict
DEYOUNG COMM	Functional as a short social handle and favicon-safe compact mark, but not strong enough to lead. Adopted as the secondary compact wordmark only.	Secondary mark

Table 1.1: Brand name option evaluation

The brand personality must communicate communication, intelligence, trust, speed, human-like interaction, professional service, and modern infrastructure. In practice this translates to a voice that is precise rather than hype-driven, an interface that shows real system state rather than decorative activity, and a visual identity built on the disciplined use of red, white, and black. The brand never claims sentience, never fabricates social proof, and never dresses a roadmap item as a shipped feature. This restraint is itself a positioning weapon in a market where most AI landing pages are interchangeable gradients and invented metrics.

Non-Negotiable Product Principles

- Honesty first: fake customers, fake reviews, fake call statistics, fake integrations, and fake system activity are forbidden. If a capability is not implemented, the UI labels it as unavailable, coming soon, or configuration required.
- The product sells communication, not AI: the user should feel they hired an employee, the caller should feel they are talking to someone who understands them, and the owner should see exactly what their AI employee is doing.
- Voice quality is a product feature: the AI must sound conversational, handle interruptions naturally, and stay silent when nothing real is happening. No beeps, no fake waiting sounds, no robotic filler.
- Consent is architectural: voice cloning, recording, transcription, and data retention all require explicit, revocable, auditable consent designed into the schema rather than bolted on.
- Free-first economics: the architecture must reach a first paying customer without sunk infrastructure cost, while never sacrificing security or reliability to claim something is free.

Deliverable Index

The master brief demands twenty-four deliverables before implementation. They are distributed across this dossier as follows: research report and competitor analysis in Chapter 2; product requirements in Chapter 3; technical specification in Chapter 4; system and worker architecture in Chapter 5; voice architecture in Chapter 6; database schema and API architecture in Chapter 7; permission, security, and consent models in Chapter 8; UX sitemap and screen inventory in Chapter 9; design system, brand identity proposal, and logo direction in Chapter 10; the 3D and motion system proposal in Chapter 11; responsive strategy and deployment architecture in Chapter 12; cost analysis and free-first strategy in Chapter 13; and the risk register and implementation roadmap in Chapter 14. Phase 2, which follows this dossier, implements the marketing website and the web application against these specifications.

2. Market Research and Competitor Analysis

Market Context

Voice AI agents crossed from demo to deployment between 2023 and 2025 as three component costs collapsed simultaneously: streaming speech-to-text fell to well under one cent per minute, streaming text-to-speech reached conversational quality at scale, and large language model inference became fast enough to sustain sub-second conversational turn-taking. Public coverage of the category in 2025 and 2026 describes a maturing market where buyers no longer ask whether AI can hold a phone conversation, but instead compare latency, interruption handling, telephony coverage, compliance posture, and price per minute. That shift matters for a new entrant: the differentiation battle has moved from existence of the technology to the quality of the employee experience around it, which is exactly where DEYOUNG COMMUNICATION's hiring metaphor, control room, and consent architecture are aimed.

At the same time, the category's pricing is publicly transparent in a way most software categories are not. Several competitors publish per-minute ranges, which allows a precise cost position to be chosen rather than guessed. The research below draws on public pricing pages, published plan structures, and community comparisons as of late 2025 through 2026, and every figure used in this dossier is labeled with its public source character so the numbers can be re-verified before any go-to-market claim is made. No private or scraped data is used, and no figure is invented.

Competitor Profiles

Platform	Positioning and strengths	Public pricing signal	Gaps DEYOUNG can attack
Bland AI	Enterprise-focused voice agent API. Strong telephony spine, pathways feature for routing, established brand with agencies.	Roughly \$0.11-0.14 per minute in tiered plans; published base fees on Build tiers around \$299 per month; telephony billed separately.	High minimum spend prices out individuals and small businesses. Hiring metaphor and human handoff tooling are thin.
Retell AI	Popular with startups building on top of it. Flexible LLM/TTS/telephony choices, good conversation quality tuning.	Commonly reported all-in ranges around \$0.07-0.31 per minute depending on chosen components; starter plans around \$29.	Costs swing widely with component choice. Buyer assembles their own employee experience; little unified control room.
Vapi	Developer-first orchestration, bring-your-own-provider model with speech, LLM, and voice billed at cost.	Platform portion around \$0.05 per minute plus provider costs at cost, per public plan pages and user reports.	Developer-centric: no turnkey hiring flow for non-technical business owners, limited customer memory and consent tooling.

Platform	Positioning and strengths	Public pricing signal	Gaps DEYOUNG can attack
Synthflow	No-code agent builder aimed at non-technical users, template library, strong funnel marketing.	Plan-based tiers reported from roughly \$29 to \$99+ with per-minute rates in the \$0.08-0.09 area on higher tiers.	Conversational depth and control-room transparency are limited; geared to volume outbound more than durable inbound employees.
Tel-Agent	Open-source (AGPL-3.0) LiveKit-based voice agent starter. Full pipeline reference including SIP bridge and agent worker.	No platform fee when self-hosted; operators pay providers and infrastructure directly.	Self-hosting burden, no multi-tenant product layer, AGPL obligations make direct code reuse unattractive commercially.

Table 2.1: Direct competitor profile summary (public information, 2025-2026)

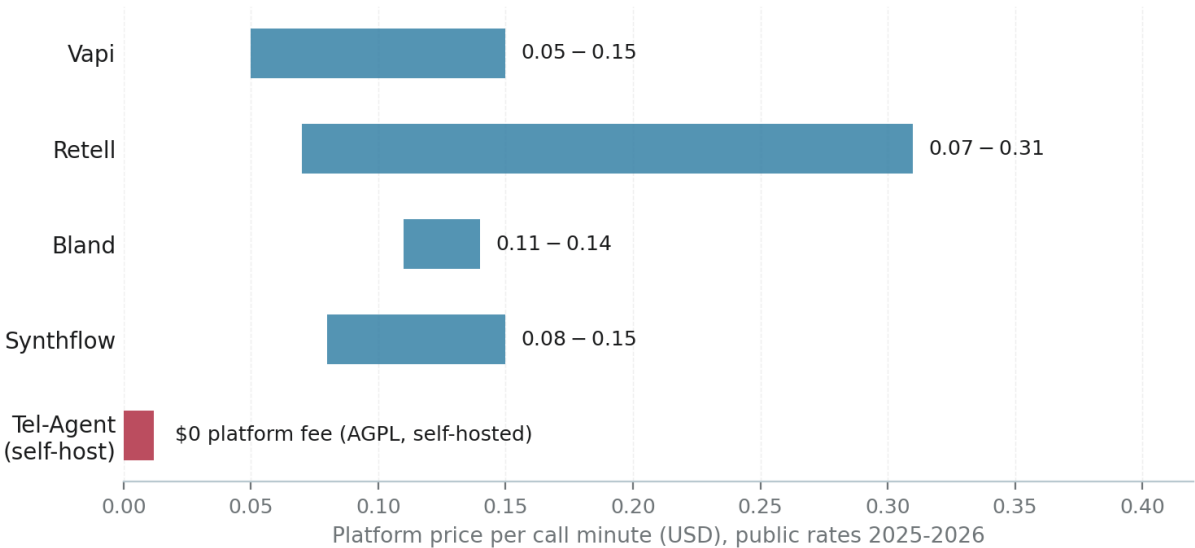


Figure 2.1: Public per-minute platform price ranges, 2025-2026. Ranges summarize public plan pages and independent comparisons; verify before external use.

The pricing picture yields a clear strategic conclusion. The hosted platforms monetize a platform margin on top of raw provider costs, and the open-source option shifts all integration and operations burden to the operator. DEYOUNG COMMUNICATION's free-first, bring-your-own-key tier can sit structurally below every hosted competitor for cost-sensitive users while the managed tier remains competitive at small-business scale, because the platform margin is decoupled from provider spend. This is only honest if the BYO tier is real: keys must be encryptable per organization, metered, and never exposed to the client, which is specified in Chapter 8.

Tel-Agent Assessment and License Analysis

Tel-Agent is an open-source project (repository github.com/Dpro-at, AGPL-3.0) that demonstrates a complete LiveKit-based voice agent pipeline: a media bridge from SIP/WebRTC into LiveKit rooms, a

Python agent worker consuming the audio stream, and pluggable speech and model providers. Its value to this project is architectural confirmation that the LiveKit agent pattern works end-to-end for telephony. Its code is deliberately not being copied.

Scenario	Consequence
Copy Tel-Agent server code into the platform	AGPL is strong network copyleft: offering the platform over a network triggers source-disclosure obligations for the entire derivative work, including proprietary orchestration and billing code. Not acceptable for a commercial SaaS.
Run Tel-Agent unmodified as an isolated self-hosted component	Still AGPL for that component. Users interacting with it over the network would be entitled to its source (which they have), but coupling it tightly with proprietary code risks derivative-work arguments. Operationally doubles the deployment surface.
Study architecture, reimplement independently	Clean-room reimplement of documented, public concepts (rooms, tracks, agent workers) carries no AGPL obligation because copyright protects expression, not ideas. This is the chosen path.
Commercial licensing	If deep reuse ever becomes valuable, the project maintainer could be approached for a commercial license. Not required under the independent-implementation strategy.

Table 2.2: AGPL-3.0 implications for DEYOUNG COMMUNICATION

Nigeria and Emerging-Market Context

The master brief calls out Nigeria accessibility as an evaluation dimension, so the research treats it as a first-class constraint rather than a footnote. Three facts shape the plan. First, WhatsApp is the dominant business messaging channel in Nigeria, and Meta has announced per-message charges for businesses on the WhatsApp Business Platform in Nigeria taking effect October 1, 2026, which means WhatsApp economics must be modeled per conversation and disclosed to users rather than advertised as free. Second, bulk SMS remains cheap and fast, which keeps SMS a viable, low-cost channel and a practical fallback where data connectivity is intermittent. Third, local infrastructure investment, including operator AI data-center expansion, is improving hosting options, though the default architecture still assumes regions outside Nigeria and optimizes for latency to African network operators where provider presence allows.

Practical consequences for the platform: pricing tiers must support naira-friendly entry points and usage-based billing rather than large fixed seats; SMS and WhatsApp channels carry country-specific cost tables surfaced honestly in the UI; and the free-first strategy in Chapter 13 deliberately uses services with generous free tiers so a Lagos agency can bring its first client onto the platform at near-zero fixed cost. Payment-provider selection is left as a configuration decision documented in the integration matrix, because local payment rails change faster than the platform should ship hard-coded opinions about them.

3. Product Requirements Document

Users and Accounts

The platform supports three account types that share one codebase and one permission model. Individual users create a single personal workspace with one or more AI assistants, useful for solo professionals such as consultants, creators, and independent service providers who miss calls while working. Business accounts create an organization with one or many AI employees, shared team inbox, customer memory, and role-based access for staff. Agency accounts manage multiple client organizations under one parent account, with cross-organization reporting, white-label touches, and per-client billing separation. The agency type is the highest-value segment because it multiplies seat-based usage across client organizations, and its needs (isolation, audit, delegated administration) drive several schema decisions in Chapter 7.

AI Employee Capabilities

An AI employee is a first-class entity the user configures as they would onboard a staff member. The builder captures name, role, purpose, personality, tone, speaking style, business instructions, conversation rules, escalation rules, knowledge sources, tool permissions, channels, working hours, human handoff rules, voice, language, security permissions, and memory behavior. Role templates (AI Receptionist, Sales Assistant, Customer Support Agent, Appointment Scheduler, Virtual Assistant, Front Desk Agent, Lead Qualification Agent, Business Information Agent, Personal Assistant) pre-fill sensible defaults but every field remains editable, because the builder must feel like configuring a real employee rather than filling a toy chatbot form.

Capability	Phase 0 (foundation)	Phase 1 (core product)	Phase 2 (scale)
AI employee builder	Full schema, UI with all fields, honest validation	Templates library, voice preview	Multi-language employees
Phone channel	UI shows Not connected until telephony configured	SIP trunk + number provisioning, inbound and authorized outbound	Regional number catalogs, carrier failover
Web chat	Embeddable widget with real transcript	Proactive triggers, visitor context	Co-browsing hooks (integration required)
SMS and WhatsApp	Channel pages with connection states	Two-way messaging with templates and opt-in capture	Campaign automation flows
Live call control room	Empty state and read-only room	Live transcript, waveform, AI action display, takeover, transfer, notes	Barge-in coaching, whisper to AI

Capability	Phase 0 (foundation)	Phase 1 (core product)	Phase 2 (scale)
Knowledge system	Upload with honest processing states	Chunking, embeddings, retrieval with citations	Versioning, refresh, CRM sync
Automations	Schema and canvas placeholder	Trigger, condition, AI action, tool action, routing, log	3D node-graph view, marketplace
Voice studio	Voice catalog with samples	Voice signature visualization, speed and prosody tuning	Consent-based cloning with revocation
Analytics	No data, honest empty states	Per-employee and per-channel metrics from real usage events	Cohort and outcome analytics

Table 3.1: Capability matrix by delivery phase

The phasing rule is that every phase ships interfaces that are real where implemented and honestly labeled where not. A channel page that has never been configured reads Not connected with the exact steps required; a dashboard with zero calls reads No calls yet and offers the action that creates the first call; an automation canvas exists only once its execution engine exists, never as a decorative mockup. This is the single most brand-defining product decision in the document, and Chapter 9's screen inventory encodes it as a required empty state per screen.

Human Handoff

Handoff is modeled as an explicit state machine: AI handling, human requested, transfer pending, human connected, human handling, returned to AI, and call ended. The control room surfaces the current state, who requested the handoff, and why. When a human takes over, the AI stops speaking immediately, the transcript continues in a shared timeline with speaker attribution, and the AI remains context-loaded so the conversation can return to it without information loss. Escalation rules on the employee decide when the AI requests a human on its own, including confidence triggers, sentiment triggers, and keyword triggers; every escalation is logged with its reason so owners can tune rules from evidence rather than vibes.

Mobile Application

Native mobile is planned as a companion, not a shrunken dashboard: Home, Calls, AI, Inbox, and More tabs oriented around monitoring and takeover on the go. App Store and Google Play entries remain honestly labeled Coming soon with a real waitlist mechanism until builds ship; download links are never faked. Mobile-specific interaction patterns (call takeover as a native call screen, push alerts for escalations, swipe actions on inbox items) are specified so the mobile app is designed, not squeezed. Performance rules from Chapter 11 apply with stricter budgets on mobile, and every 3D element falls back to its CSS tier on low-end devices.

4. Technical Specification

Stack Selection

The stack was selected against eight explicit criteria from the master brief: cost, latency, reliability, developer experience, Nigeria accessibility, free-tier availability, scalability, and vendor lock-in, plus privacy and licensing as gating constraints. The chosen stack keeps every component either open-source or available with a meaningful free tier, and it avoids any provider whose free tier cannot carry a first customer.

Layer	Choice	Rationale
Frontend	Next.js (App Router) + TypeScript + Tailwind CSS + shadcn/ui	One framework covers marketing site, web app, and embeddable chat widget with code-splitting for 3D bundles. TypeScript is non-negotiable for a product with this much state. shadcn/ui gives accessible primitives that are owned, not rented.
API layer	Next.js route handlers (REST) + WebSocket service for realtime	Colocated REST keeps Phase 0 deployment simple and reviewable. Realtime transcript and call events use a dedicated WebSocket/SSE service so page interactions never compete with call streams.
Database	PostgreSQL (Neon or self-hosted) + Prisma ORM	Relational integrity for organizations, consent, and audit is a compliance requirement, not a preference. Row-level scoping enforced at the query layer. Prisma migrations are reviewable SQL.
Cache and queues	Redis (Upstash) for cache and streams	Free tier carries the first customers; streams power worker queues with consumer groups; pub/sub fans out live call events.
Object storage	S3-compatible (Cloudflare R2)	Zero-egress pricing matters for audio recordings and knowledge documents; standard S3 API avoids lock-in.
Voice media	LiveKit (open-source) with SIP bridge, Twilio Programmable Voice as alternate ingress	LiveKit agents pattern is proven by open-source reference projects; SIP bridge terminates real phone calls; Twilio option covers regions where its numbers are easier to provision.
AI providers	Deepgram or Whisper for STT; OpenAI or Anthropic for LLM; ElevenLabs or PlayHT for TTS behind a provider abstraction	Each slot has two candidates behind one interface, so provider pricing or policy changes never require product surgery. BYO keys supported with encrypted storage.
Auth	Self-hosted session auth with argon2 password hashing, or Auth.js where a provider is required	Session cookies keep auth free-tier friendly; RBAC is enforced server-side in every query path; no client-side tokens for privileged operations.

Table 4.1: Stack decisions and rationale

Provider Abstraction

Every external capability (STT, LLM, TTS, telephony, storage, messaging) sits behind an internal interface with at least two implementations. The abstraction defines latency budgets, streaming semantics, error contracts, and cost meters per provider so the orchestrator can route by policy: cheapest, fastest, or pinned by the account owner. Provider credentials live server-side only, encrypted per organization for BYO keys, and are never included in any client bundle. This single design decision is what makes the free-first and BYO strategies honest instead of aspirational.

Realtime Transport

Voice media flows through LiveKit rooms; call control events, transcript deltas, and AI action notifications flow over a WebSocket gateway backed by Redis pub/sub, so any dashboard session can subscribe to the calls it is permitted to see. Web chat uses SSE for visitor-to-server messages and the same WebSocket for agent-side views. All realtime payloads are typed and versioned, because a live call control room cannot tolerate schema drift between the worker that emits events and the client that renders them.

5. System and Worker Architecture

Request Lifecycle

Every conversation, inbound or outbound, follows one lifecycle: channel ingress (phone call, chat message, SMS, webhook), gateway (media gateway for audio, API gateway for events), the AI orchestrator, which resolves employee configuration, permissions, and context policy; context and memory assembly from organization-scoped stores; worker routing to the specialist workers; external provider calls where required; response streaming back to the channel; and usage events written for metering, audit, and analytics. The orchestrator is deliberately thin: it decides who works, with what permissions, on what data, and when to escalate. It does not itself transform audio or generate text, which keeps the brain auditable and the workers replaceable.

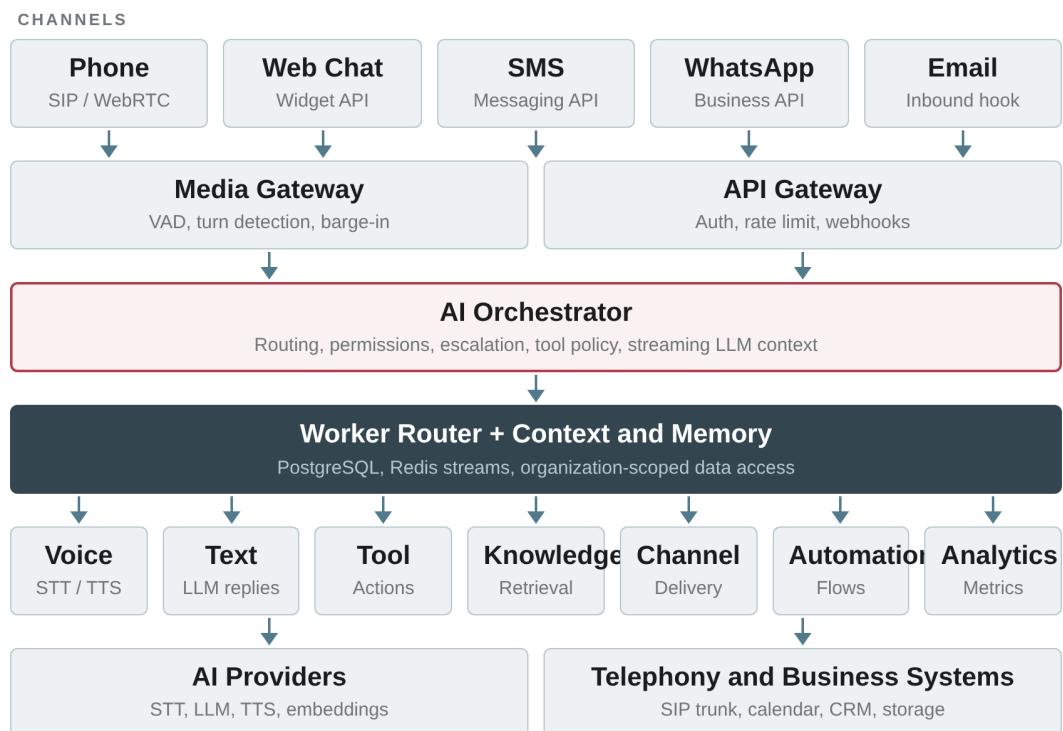


Figure 5.1: Platform architecture overview. Layers connect top to bottom; the orchestrator (red border) is the only component that makes policy decisions.

Worker Topology

Worker	Responsibilities	Scale and idle policy
Voice worker	Owns the live audio session: VAD, turn detection, barge-in, streaming STT and TTS bridging, interruption cuts.	One per active call; scales with concurrent calls; idles at zero when no calls are live.
Text worker	Streams LLM responses for chat, SMS, WhatsApp, and email with per-employee prompt and rules.	Pool scales with inbound message rate; cold-start budget under two seconds.
Tool worker	Executes approved tools (calendar, CRM writes, lookups) in a sandbox with per-tool permission checks and timeouts.	Queue-driven; hard timeouts per tool; every execution logged with arguments and result.
Knowledge worker	Ingests documents (chunk, embed, index) and serves retrieval with citations for answer grounding.	Ingestion burst-scaled; retrieval path cached; processing states surfaced honestly to the UI.
Channel worker	Delivers messages to each connected channel and normalizes inbound events to the internal envelope format.	Per-channel queues; backpressure surfaced as channel status (Connected, Needs attention).
Automation worker	Evaluates triggers, runs conditions, sequences AI and tool actions, writes run logs.	Event-driven; retries with exponential backoff; dead-letter queue with admin visibility.

Worker	Responsibilities	Scale and idle policy
Memory worker	Maintains unified customer profiles; merges interactions, extracts preferences, respects retention rules.	Batch plus event-driven; every write consent-aware and organization-scoped.
Analytics worker	Aggregates usage events into per-employee, per-channel, and outcome metrics.	Scheduled rollups; dashboards read pre-aggregated tables, never raw events in request paths.

Table 5.1: Worker responsibilities and scale policy

Orchestration Decisions

The orchestrator makes five decision classes on every turn: which worker is needed (a pure routing decision), when to start and stop it (idle shutdown is the default, so an organization with no traffic consumes no worker compute), which provider to use (policy-driven by cost, latency, and pinning), what permissions apply (the worker runs with a scoped token that encodes what data it may touch), and when to escalate to a human (escalation rules plus confidence signals). Every decision is written to the audit log with its inputs, because Chapter 8 requires that a compliance officer can reconstruct why any AI action happened. Workers are independently scalable and independently replaceable, which is the property that lets the platform add a channel or swap a provider without touching the brain.

6. Voice Architecture

Realtime Pipeline

The voice pipeline is a streaming chain with no batch steps: phone or WebRTC audio enters the media gateway; voice activity detection segments speech; streaming STT emits partial hypotheses that feed turn detection; the turn boundary triggers context assembly and the LLM with response streaming; TTS begins speaking the first sentences while generation continues; and audio flows back to the caller. Tool execution slots into the chain as an inline step with a natural filler phrase approved in the employee's voice profile, for example acknowledging the wait before checking a calendar, and the tool's real result drives the next spoken sentence. Nothing in the chain is permitted to fake a result or insert synthetic waiting audio.

Stage	Budget	Notes
VAD turn end detection	80 ms	Energy plus semantic turn classifier; endpointing tuned per employee pace.
STT final hypothesis	200-300 ms	Streaming STT partials already available; final at turn end.

Stage	Budget	Notes
Context and memory fetch	60 ms	Pre-fetched on call start; Redis-cached customer profile.
LLM first token	300-500 ms	Prompt sized for first-token speed; longer reasoning deferred to later sentences.
TTS first audio	120-200 ms	Streaming synthesis begins at first sentence boundary.
Total turn response	760-1140 ms	Comfortable conversational window; sub-second perceived with early TTS onset.

Table 6.1: End-to-end latency budget (target, p70)

Turn-Taking and Interruption

Interruption handling is a safety-gated behavior, not a random feature. While the AI speaks, VAD continues listening; caller speech crossing a threshold immediately fades AI audio out within 150 ms, aborts pending TTS, and marks the generation as interrupted so the LLM receives the new utterance with conversational context. The AI may itself interrupt the customer only under gated conditions: urgent correction (the customer states something factually wrong that the business rules flag), clarification that prevents a misunderstanding, or acknowledgement-style backchannels that humans naturally use. Every AI-initiated interruption is logged with its gate classification, because an AI that constantly interrupts is a defect, and the log is how the behavior gets tuned.

Naturalness Standards

The voice must not sound like an assistant reading a script. Standards applied in tuning and QA: prosody varies with sentence type; pauses follow punctuation and thought boundaries; short acknowledgements are used the way humans use them; speaking speed adapts to the caller's measured rate; hesitation appears only where a human would hesitate, for example before delivering a number that was just looked up. Forbidden by policy: unnecessary beeps, fake hold music, please wait filler while nothing is being checked, confirmation of every sentence, monotone delivery, and dead air while a tool executes. When a tool runs, the employee says something true about it, then reports the real result.

Voice Cloning and Consent Gates

Voice cloning is optional, permission-gated, and revocable by design. The consent record stores the voice owner, authorization status, consent timestamp, allowed usage scope, voice ID, creation date, revocation state, and full audit history. Cloning cannot be initiated without an explicit consent artifact (recorded attestation or signed document); usage scope limits where the cloned voice may be deployed; and revocation is immediate, removing the voice from all employees and blocking any in-flight synthesis. The schema in Chapter 7 treats consent as data lineage: a voice without a valid consent record is not loadable by the TTS worker, enforced at the query layer rather than by UI convention.

7. Database Schema and API Architecture

Core Schema

The schema is PostgreSQL with organization-scoped rows as the isolation primitive. Every business table carries `organization_id`, and server-side query helpers refuse unscoped reads, so cross-tenant leakage is a code-review problem with a compile-time ally rather than a hope. The table below lists the core entities; column-level detail lives in the Prisma migration files that Phase 2 generates from this specification.

Entity	Key columns and semantics	Notable constraints
users, organizations, memberships	users (id, email, password_hash, status); organizations (id, type, name, settings); memberships (role: owner, admin, operator, viewer)	Agency parent linking via <code>organizations.parent_id</code> ; role checked on every mutation.
ai_employees	id, org, name, role, purpose, personality, tone, instructions, rules, escalation, channels, hours, voice_id, language, memory_policy, status	Versioned config; draft vs deployed states; deletion soft with retention clock.
conversations, messages	conversations (id, org, channel, customer, employee, status); messages (id, conversation, direction, content, speaker, metadata)	Full-text search on transcripts; retention policy enforced by scheduled job.
calls, transcripts, events	calls (id, org, direction, from, to, employee, duration, outcome, recording_ref); transcript rows (speaker, text, confidence); call events (state machine transitions)	Recording gated by consent flags; waveform amplitude series stored for live UI.
customers	id, org, name, phone, email, company, tags, preferences, consent_status, merged_into	Unified profile; dedupe by phone/email; consent flags drive AI access scope.
knowledge_sources, chunks	sources (id, org, type: pdf, docx, txt, url, faq, manual; status, version); chunks (id, source, content, embedding, citation)	Processing states honest: pending, processing, indexed, failed; delete cascades chunks and vectors.
automations, runs, steps	automations (id, org, trigger, conditions, actions graph); runs (id, automation, status, log); steps (id, run, type, input, output, error)	Every run auditable end to end; dead-letter status surfaced in UI.
voices, voice_consent	voices (id, org, provider_voice_id, name, gender, language, sample_ref); consents (id, voice, owner, scope, authorized_at, revoked_at, artifact_ref)	TTS worker loads voices only where an unrevoked consent row exists.
integrations, credentials	integrations (id, org, type, status, config); credentials (id, org, type, encrypted_value, key_version, last_rotated)	Envelope encryption; decrypted only inside the tool worker; never sent to client.

Entity	Key columns and semantics	Notable constraints
phone_numbers, channels	phone_numbers (id, org, number, provider, capabilities, status); channel connections per org with state and last_error	States: connected, needs_attention, not_connected, coming_soon.
usage_events, audit_logs	usage (id, org, kind, units, cost_estimate, refs); audit (id, org, actor, action, target, reason, ip, at)	Immutable append-only tables; feeds billing, metering, and compliance review.
feature_flags, waitlist	flags (key, org scope, enabled); waitlist (id, email, platform, source, created_at)	Honest gating of coming-soon surfaces; real waitlist storage for mobile apps.

Table 7.1: Core entities

API Surface

Resource	Methods	Semantics
/api/v1/auth	POST	Session login, logout, refresh; rate-limited; argon2 hashing; audit-logged.
/api/v1/organizations	GET, POST, PATCH	Create and manage orgs; agency parent-child linking; settings including data retention defaults.
/api/v1/ai-employees	GET, POST, PATCH, DELETE	CRUD plus deploy and version endpoints; draft/deploy separation; per-field validation.
/api/v1/conversations, calls	GET, POST	List with filters; call control actions (takeover, transfer, end) validated against state machine.
/api/v1/customers	GET, POST, PATCH	Unified profiles; merge endpoint; consent flag updates restricted to admin roles.
/api/v1/knowledge	GET, POST, DELETE	Source upload with processing state; refresh; delete cascades index entries.
/api/v1/automations	GET, POST, PATCH, DELETE	Graph validation on save; run inspection endpoints.
/api/v1/voices	GET, POST	Catalog with samples; cloning initiation gated on consent artifact upload.
/api/v1/integrations	GET, POST, DELETE	Connect flows store encrypted credentials; status endpoints reflect real provider checks.
/api/v1/usage, audit	GET	Metering and audit queries; admin scopes; export formats for compliance.

Resource	Methods	Semantics
/api/v1/webhooks	POST	Inbound provider events; signature-verified; replay-protected; normalized to internal events.
/ws/realtime	WS	Typed, versioned events: transcript deltas, AI action state, call state, channel status.

Table 7.2: Primary REST surface (v1)

Events and Webhooks

Internally, every meaningful state change emits a typed event onto Redis streams: conversation started, turn completed, tool executed, escalation requested, handoff completed, automation run finished, usage recorded. Workers consume these streams with consumer groups, which gives at-least-once delivery with idempotent handlers. Externally, organizations can register outbound webhooks for a subset of events, with signature verification, retry with backoff, and a delivery log visible in the UI. Outbound webhooks are an integration surface for real businesses, so they behave like production infrastructure: replay protection, per-endpoint secrets, and honest failure states rather than silent drops.

8. Permission, Security and Consent Models

Role-Based Access Control

Access control is organization-scoped RBAC with four roles, enforced server-side on every query and mutation. The UI renders what the API permits; nothing is hidden client-side that the server would still accept. Agency accounts add delegated administration: an agency admin can act inside client organizations only through an explicit agency grant, and every such cross-org action is audit-logged with both the acting user and the target organization.

Capability	Owner	Admin	Operator	Viewer
Manage billing and plan	Yes	No	No	No
Manage roles and membership	Yes	Yes	No	No
Create and deploy AI employees	Yes	Yes	No	No
Edit employee configuration	Yes	Yes	Drafts only	No
Take over live conversations	Yes	Yes	Yes	No
View conversations and transcripts	Yes	Yes	Yes	Yes

Capability	Owner	Admin	Operator	Viewer
Manage knowledge and integrations	Yes	Yes	No	No
Manage voice consents (grant, revoke)	Yes	Yes	No	No
Export audit and usage data	Yes	Yes	No	No

Table 8.1: Role permission matrix

Security Model

Security is treated as a product feature with a visible surface, not a checkbox list. The implemented baseline: session authentication with hardened cookies and CSRF tokens on state-changing routes; argon2id password hashing with per-user salts; rate limiting on auth, API, and webhook endpoints; input validation with typed schemas at every boundary; parameterized queries only, via the ORM, with organization scoping in query helpers; encrypted secrets using envelope encryption with per-environment keys, decrypted only inside the worker that needs them; secure file uploads with type and size validation, virus-scan hook, and randomized storage keys; strict CSP with no inline script execution; and dependency scanning in CI. Provider secrets never reach the client bundle, verified by a build-time check that greps the client output for secret variable names.

- Organization isolation is enforced in query helpers, so an unscoped read is a code path that cannot compile, not a convention reviewers must remember.
- All privileged actions write immutable audit entries: actor, action, target, reason, source IP, timestamp.
- Webhook endpoints verify signatures, reject replays using timestamp windows and nonces, and normalize payloads before they touch domain logic.
- Rate limits and lockout policies are conservative by default and tunable per organization by owners.
- Secrets rotation is a first-class operation with key versions, so rotation does not require credential re-entry by every user.

Consent Model

Consent is modeled as typed, revocable artifacts rather than a checkbox bit. The platform tracks recording consent (per call, per channel, per jurisdiction notice), transcription consent, data retention consent, voice cloning consent, and channel opt-in state (SMS and WhatsApp opt-ins with timestamps and source). Each artifact records who consented, in what form (verbal attestation captured in the transcript, written acceptance, digital signature), when, for what scope, and its current revocation state. Revocation effects are immediate and mechanical: a revoked recording consent deletes stored audio and

keeps only the transcript if separately consented; a revoked cloning consent blocks the TTS worker from loading that voice; a channel opt-out stops all outbound messaging through that channel. AI disclosure requirements are configurable per employee and per jurisdiction, so the platform can honestly say an AI is speaking where the law or the business requires it.

Privacy and Retention

Privacy pages and controls are generated from the same rules the engine enforces, so documentation cannot drift from behavior. Retention defaults are conservative, per-data-class, and owner-adjustable within policy ceilings; deletion requests execute across relational rows, object storage, vector indexes, and analytics rollups in one auditable job with a completion receipt. Export gives users their data in documented formats. Human access to recordings and transcripts is itself logged, because insider access is part of the privacy surface an enterprise buyer audits. Third-party provider disclosure lists every sub-processor by function with data classes shared, kept current in the integration registry rather than hand-maintained on a marketing page.

9. UX Sitemap and Screen Inventory

Marketing Sitemap

Group	Routes
Core	/ (homepage with hero product treatment), /product, /voice-studio, /developers, /pricing, /security, /about, /contact
Product pages	/product/ai-employees, /product/voice, /product/conversations, /product/channels, /product/knowledge, /product/automations, /product/integrations
Solutions	/solutions, /solutions/individuals, /solutions/business, /solutions/agencies
Resources	/resources, /resources/guides, /resources/documentation, /resources/changelog
Account	/login, /signup
Legal	/privacy, /terms, /cookies, /acceptable-use, /voice-consent, /data-retention

Table 9.1: Public website structure

Application Navigation

The app sidebar carries eleven sections: Overview, AI Employees, Live Calls, Inbox, Customers, Knowledge, Voice Studio, Automations, Integrations, Analytics, Settings. Each section has a purpose stated in one line so the IA stays legible as features grow: Overview is the honest state of the account; AI

Employees is hiring and configuration; Live Calls is monitoring and control; Inbox is every conversation across channels; Customers is memory; Knowledge is what the AI is allowed to know; Voice Studio is how it sounds; Automations is what it does on its own; Integrations is what it is connected to; Analytics is what actually happened; Settings is who can do what.

Screen Inventory with Required Empty States

Screen	Primary content	Required honest empty state
Overview dashboard	Account status cards: employees, live calls, conversations, channel health, usage.	New account: Create your first AI employee, Connect an integration, No calls yet. Zero invented metrics.
AI Employees list	Cards with name, role, status, channels, last activity.	No employees: Create your first AI employee with role templates.
Employee builder	Sections for identity, instructions, knowledge, tools, channels, voice, permissions.	Fields validate honestly; unavailable capabilities render labeled Coming soon or Configuration required, never disabled-fake.
Live Calls room	Active call cards; detail view with transcript, waveform, AI action, controls.	No live calls: No live calls right now, with link to test call (only if telephony configured; otherwise states Not connected).
Inbox	Conversation threads across channels with assignment and status.	No conversations: Nothing here yet. Connect a channel to start receiving conversations.
Customers	Profile cards with history, notes, tags, consent status.	No customers: Customers appear here after your first conversations.
Knowledge	Source table with type, status, version, last updated.	No sources: Add your first knowledge source (PDF, DOCX, TXT, URL, FAQ, or manual entry).
Voice Studio	Voice catalog with samples, tuning controls, consent management.	No custom voices: Browse standard voices or add a consented custom voice.
Automations	Flow list with run history.	No automations: Build your first automation from a real event trigger.
Integrations	Integration cards with connection states.	Each card states Connected, Needs attention, Not connected, or Coming soon with the actual reason.
Analytics	Metric pages driven only by real usage events.	Not enough data yet: Analytics populate as your AI employees handle real conversations.

Screen	Primary content	Required honest empty state
Settings	Organization, members and roles, security, retention, billing.	New org defaults with explicit review prompts for retention and access policies.

Table 9.2: App screen inventory (selected core screens)

Onboarding Flow

Onboarding is sequenced to reach an honest first win: create an organization, create the first AI employee from a role template, add one knowledge source (or skip with the employee honestly reporting it has no business knowledge yet), choose a standard voice with a real preview, run a web chat test conversation, and only then decide whether to connect telephony or messaging. Every step shows what is real: a test chat works immediately because the chat channel is native, while a phone number is presented with its actual configuration requirements and regional availability. The flow never fabricates a demo call recording to simulate progress; it uses a clearly labeled simulated practice mode only for voice, marked Practice mode: simulated audio, so the user can hear pacing and interruption behavior without a provisioned number.

10. Design System, Brand and Logo Direction

Color Tokens

Token	Value	Usage rules
brand.red	#E10600	Primary CTAs, active states, selected navigation, waveform accents, live indicators. Intentional, never decorative fills of large areas.
brand.black.deep	#090909	Premium dark sections, navigation, footer, product chrome, deep contrast bands.
brand.black.soft	#111111	Card surfaces on dark, hover states on dark chrome.
brand.white	#FFFFFF	Content areas, cards on light, forms, product explanations.
brand.white.off	#F5F5F3	Light section backgrounds, alternating bands, table zebra on light.
brand.gray.muted	#A1A1A1	Secondary text, captions, inactive states on dark.
brand.gray.border	#262626	Borders and dividers on dark surfaces.

Token	Value	Usage rules
semantic.status	Derived neutrals plus brand red for live/urgent; green and amber tuned to low saturation for success and warning.	Status is never communicated by color alone; every state pairs an icon or label.

Table 10.1: Core color tokens (RED / WHITE / BLACK system)

The red is strategic, never atmospheric. It appears where the product is alive: a live call indicator, the waveform peak, the primary action. Black carries structure and seriousness; white carries clarity. Explicitly banned: purple and blue AI gradients, neon cyberpunk styling, glassmorphism panels, rainbow or red-on-red gradients, cartoonish AI imagery, emoji icons, generic robot illustration, floating blobs, and pill-shaped everything. Neutral hierarchy is built from the black and white axis, so the palette stays three-voiced: red speaks, black structures, white clarifies.

Typography

One display face and one interface face, both justified by function. Display: a geometric-intermediate sans with strong caps and tight tracking for headlines and the wordmark, giving the confidence of a network operator rather than a startup template. Interface: a highly readable UI sans with excellent numerals and tabular figures for dashboards, transcripts, and durations. Both are loaded as two weights each (regular, bold) plus one mono for transcripts and system events, because transcript reading benefits from a mono rhythm that visually separates machine-logged events from speech. Type scale runs from 64px display down to a 13px floor for meta text, with spacing tokens in a 4px base grid.

Logo Direction

The mark concept is a speech-act monogram: the DEYOUNG D constructed from two offset vertical bars and a horizontal voice bar, forming a letterform that reads as both a D and a leveled waveform in a single glance. It deliberately avoids the AI clichés (no brain, robot, circuit, infinity, gradient orb). The system delivers: primary lockup (monogram plus DEYOUNG COMMUNICATION wordmark, left-aligned), compact mark (monogram alone for favicon and app icon), wordmark alone for co-branding contexts, monochrome versions in pure black and pure white, light and dark variants with adjusted bar weights for equal optical mass, and a 3D-extrudable version using clean rectangular geometry with no fine detail, specified in Chapter 11's material tokens. The wordmark sets DEYOUNG in the display face with COMMUNICATION in tracked caps below, establishing the two-line hierarchy used across the site and product chrome.

Component Standards

System	Standard
Buttons	Red primary with white text; black secondary; ghost tertiary; 2px radius (sharp, not pill); loading states replace label text with real operation state; disabled requires a tooltip reason.
Forms	White surfaces, 1px neutral borders, red only for focus and error; validation messages explain the fix; labels always visible, never placeholder-only.
Status	Five states across the product: live (red pulse plus label), needs attention (amber plus icon), connected (green plus check), not connected (neutral), coming soon (dashed border plus label). Never color-only.
Tables	Dense professional grid: sticky headers, zebra with off-white, mono numerals right-aligned, row actions in a fixed trailing column.
Navigation	Dark chrome with white text; active section marked by red left rule and white text; muted gray for inactive; keyboard-accessible with visible focus.
Toasts and modals	Toasts for operation outcomes with real success or failure; modals reserved for destructive or consent-gated actions with explicit consequences written out.
Iconography	One professional icon set (stroke-based, 1.5px), no emoji anywhere in UI chrome.

Table 10.2: Component system highlights

11. 3D and Motion System Proposal

Philosophy

3D exists to make the product's real-time, physical nature legible: a live voice conversation, a running automation, the path of a request through workers. Every animated element must map to something the product is actually doing, driven by typed data streams from real backend state. If a scene cannot be bound to real data, it does not ship. Red is the only color permitted to move with intensity; black and white stay structural and static. Motion should feel mechanical and precise, like calibrated hardware, with a single fixed key light and soft fill consistent across every scene so the brand reads as one studio setup.

Scene Plan

Scene	Data binding and behavior	Placement
Homepage hero	Physically modeled phone plus desktop pairing showing a live call UI, built from actual product components, rendered at slight tilt with cursor parallax; no spinning, no idle loops.	/ homepage only
Live call waveform	True 3D waveform mesh driven by real amplitude and frequency data of the active or most recent call; idle renders as a flat, calm baseline.	Live Calls room, Voice Studio preview

Scene	Data binding and behavior	Placement
Voice signature	Spectral visualization generated from an actual voice sample, used to differentiate voices at a glance.	Voice Studio catalog
Worker topology diagram	Isometric system diagram of the real worker architecture, rotatable, with the option to highlight the live path of a request.	/developers, /product pages
Automation canvas (optional view)	Isometric node-graph for complex automations, switchable to standard 2D flowchart.	Automations section (Phase 2)
Logo sting	Sub-1-second extrusion and assembly of the wordmark for loading states; skippable, never blocking.	App boot, brand video

Table 11.1: Approved 3D scenes

Explicitly not using 3D: marketing feature grids, testimonials, pricing tables, and any element whose purpose is to look futuristic rather than explain the product. These stay flat, typographic, and fast. This restraint is a stated design rule, not a preference, and it is on the Section 35-style audit checklist for every page review.

Performance Budget and Degradation

Tier	Definition	Budget
Full 3D (WebGL)	Capable desktop and modern mobile.	60 fps target on mid-tier hardware; hero scene under 150k triangles and 40 draw calls; waveform under 60k triangles; bundle code-split and lazy-loaded.
Reduced 3D	Mid-tier mobile; fewer particles and capped frame rate.	30 fps floor, half the mesh budget, no post-processing.
Static and CSS fallback	Low-end devices, prefers-reduced-motion, WebGL unavailable, first paint before 3D bundle loads.	Same information, zero motion: CSS waveform bars with real amplitude data, SVG worker diagram, static hero composition. Page fully interactive before 3D loads.

Table 11.2: Rendering tiers and budgets

Data binding is enforced architecturally: no 3D component may own fake internal state used only to animate itself; every animated value subscribes to a typed stream (call amplitude series, worker activity events, automation run state). The idle rule is absolute: if nothing real is happening, the element is visibly still. Accessibility requires prefers-reduced-motion to disable camera movement, auto-rotation, and looping while preserving data-driven changes (the waveform still reflects real audio, without flourish), and every scene ships a text or ARIA-labeled equivalent describing what it shows. Motion tokens: mechanical easing with sharp attack and precise settle, no bounce or elastic curves; maximum idle

animation is none.

12. Responsive and Deployment Strategy

Responsive Strategy

Range	Marketing site	Application
1280px and up	Full hero pairing, three-column feature systems, wide tables.	Sidebar plus three-pane rooms (list, detail, context).
1024-1279px	Hero simplifies to phone-forward composition; two-column features.	Collapsible sidebar; two panes with context drawer.
768-1023px	Stacked sections; diagram switches to SVG fallback tier.	Icon rail navigation; single pane with master-detail switching.
Under 768px	Single column; all 3D drops to static or CSS tier; tap targets 44px.	Bottom tab navigation on Home, Calls, AI, Inbox, More patterns; takeover flows designed mobile-first.

Table 12.1: Breakpoints and layout behavior

The web app is designed so that monitoring and takeover work well on a phone, because that is what an owner actually does when an escalation alert fires at dinner. The future native app reuses these interaction patterns rather than shrinking the desktop. Mobile performance rules apply the Chapter 11 reduced budgets by default on small viewports, and every form validates inline with inputs sized to avoid OS zoom jumps.

Deployment Architecture

Phase 0 deployment keeps the entire platform on free-tier-capable infrastructure: the Next.js site and API on a serverless host with hobby-tier pricing; PostgreSQL on a free-tier managed Postgres; Redis on a free-tier serverless Redis; object storage on zero-egress S3-compatible storage; and the voice media layer on LiveKit Cloud's entry tier or a self-hosted LiveKit node when a SIP trunk demands it. Workers run as small containers on a host with scale-to-zero where the platform allows, or as scheduled functions in the serverless path. The deployment unit boundaries match the worker boundaries, so each worker can be promoted to dedicated hosting as load justifies it without code changes.

- Environments: production, staging (isolated data, same provider configuration shape), and ephemeral preview deployments per branch.
- CI: type-check, lint, unit and integration tests, dependency audit, client-bundle secret scan, Playwright smoke suite on marketing and app shells, and Lighthouse budget gates on key pages.

- Database changes ship only via reviewed migrations; migrations are reversible or paired with a rollback plan.
- Observability from day one: structured logs, per-worker error rates, provider latency histograms, and an honest system status page that reflects real probes.
- Backups: daily Postgres snapshots with documented restore drills; object storage versioning on recordings and knowledge documents.

13. Cost Analysis and Free-First Strategy

Provider Cost Structure

Component	Typical range	Notes
Streaming STT	\$0.005-0.01 / min	Deepgram Nova-class streaming rates publicly listed around \$0.0048-0.0092 per minute depending on channel and plan.
LLM (conversational)	\$0.005-0.05 / min	Depends on model tier and prompt size; first-token speed favors mid-tier models with streaming.
Streaming TTS	\$0.01-0.10 / min	Ranges from efficient neural voices to premium cloned voices; per-provider plan tiers swing this widely.
Telephony	\$0.005-0.03 / min	SIP trunk termination varies by destination; Nigerian routes and messaging priced separately.
Platform margin	Set by tier	Zero on BYO-keys tier; usage-billed on managed tier; never charged on provider costs the customer pays directly.

Table 13.1: Variable cost components per call minute (public list rates, late 2025-2026, order of magnitude)

These figures are planning magnitudes from public pricing pages and must be re-verified before any public claim, which is why the pricing page itself will present estimates with their basis rather than false precision. The structural point stands regardless of rate drift: provider cost is the floor, and the platform's honest differentiation is decoupling its own margin from that floor.

Infrastructure Cost Tiers

Stage	Monthly fixed cost	What it buys
Development (Phase 0)	\$0-5	Serverless hosting free tier, free-tier Postgres and Redis, zero-egress storage, preview deployments.

Stage	Monthly fixed cost	What it buys
First customers (Phase 1)	\$15-45	Paid serverless tier, small always-on worker host for call reliability, LiveKit entry tier, backups.
Growth (Phase 2)	\$150-400	Dedicated worker autoscaling, managed Postgres with read replica, multi-region SIP capacity, observability stack.

Table 13.2: Fixed infrastructure cost by stage

Free-First Architecture Decisions

The free-first strategy is a set of architectural commitments, not a discount: open-source core (Next.js, LiveKit, Postgres, Redis, Prisma); provider abstraction with BYO keys so an account can run entirely on its own provider spend with zero platform fee; idle shutdown so an organization with no traffic consumes no worker compute; queue-based activation so bursts degrade gracefully instead of billing for standing capacity; caching of knowledge retrieval and customer profiles to cut repeated provider calls; and usage-based activation so every metered thing is visible in the usage ledger before it bills. Tiers are honestly separated as FREE (BYO keys, all product surfaces, real limits), LOW COST (managed providers with usage billing), PAID (team features, automations at scale), and ENTERPRISE (SSO, audit export, custom retention, regional data residency options). Security and reliability are never traded to claim something is free: the free tier uses the same encryption, isolation, and audit paths as paid tiers.

14. Risk Register and Implementation Roadmap

Risk Register

Risk	Likelihood	Impact	Mitigation
Voice latency regresses beyond conversational budget	Medium	High	Per-stage latency histograms in production; provider routing by latency policy; prompt sizing reviews; early-TTS onset preserved in the pipeline contract.
Telephony provisioning blocks first customers	Medium	High	Twilio alternate ingress; regional number catalog with honest availability; web chat ships as the immediately working channel so onboarding never blocks on telephony.
WhatsApp per-message economics erode margins (Nigeria charging from Oct 2026)	High	Medium	Per-conversation cost model surfaced in UI; SMS fallback; template and session types routed by cost policy.

Risk	Likelihood	Impact	Mitigation
AGPL contamination from reference projects	Low	High	Independent implementation policy; CI license scan on dependencies; documented clean-room process for studied architectures.
Consent or recording compliance gap in a target jurisdiction	Medium	High	Consent artifacts as schema-level lineage; per-jurisdiction disclosure config; audit exports; legal review checkpoint before telephony launch per market.
Provider policy or price change (STT, LLM, TTS)	High	Medium	Two providers per slot behind one interface; BYO keys; usage ledger makes cost drift visible to owners immediately.
LiveKit Cloud tier limits at growth	Medium	Medium	Self-hosted LiveKit as scale path; media layer isolated behind the media gateway so swap is deployment-level.
Honest-empty-state product reads as unfinished to skeptics	Medium	Medium	Copy and design investment in empty states as onboarding moments; Practice mode for voice; docs that explain the honesty principle.
Abuse of outbound calling capability	Medium	High	Authorized outbound only with consent and rate ceilings; abuse queue in admin; feature-flagged rollout; regional compliance gates.
Solo-founder delivery bandwidth	High	Medium	Phase gates in this roadmap; P0 scope discipline; automated QA gates in CI to hold quality while velocity is limited.

Table 14.1: Top risks with mitigations

Implementation Roadmap

Phase	Scope	Exit criteria
Phase 0: Foundation (now)	Brand system and design tokens; marketing site core pages; auth; organization and employee schema; app shell with eleven sections and honest empty states; web chat employee with real LLM behind provider abstraction; audit log and usage events live.	A signup can create an org, build an employee, and hold a real web chat conversation with transcript persistence and audit entries.
Phase 1: Voice core	LiveKit media path; SIP trunk inbound; streaming STT and TTS; turn detection and barge-in; live call control room with real waveform and transcript; knowledge ingestion and retrieval; SMS channel; takeover flows.	A real phone call is answered, interrupted, escalated, and taken over by a human, with all events logged.
Phase 2: Scale product	WhatsApp channel with consented opt-in; automations engine and canvas; voice studio with consented cloning and revocation; analytics rollups; integrations registry with encrypted BYO keys; agency multi-org features.	Agency onboards a client end to end; automation runs from a real trigger to a logged outcome.

Phase	Scope	Exit criteria
Phase 3: Mobile and market	Native mobile companion (waitlist converts to TestFlight and Play tracks); regional telephony expansion; pricing page with verified cost basis; compliance pack (policy pages generated from enforced rules).	Mobile app ships with takeover on the go; at least one paying organization per tier.

Table 14.2: Phase plan (maps to the 19-phase master process)

Immediate Next Steps

With this dossier approved as the planning baseline, Phase 2 begins with the foundation build: implement the design tokens and component library against Chapter 10, stand up the marketing core pages with the hero treatment from Chapter 11, then the application shell with authentication, the organization and employee schema, and the eleven-section navigation with the honest empty states specified in Chapter 9. The first vertical slice to make real is the web chat employee, because it exercises the orchestrator, provider abstraction, transcript persistence, usage events, and audit logging end to end without waiting on telephony provisioning. Each subsequent phase then follows the exit criteria above, with the quality bar of the master brief applied at every gate: the product must feel like a real communications company to an investor, an enterprise buyer, and an engineer on first look, and it must never fake the evidence.